

2024



Synthesia Makes AI Video Production Effortless with Generative AI on AWS

Learn how Synthesia simplifies video production using NVIDIA GPU-powered Amazon EC2 Instances and PyTorch on AWS.

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Reduced ML model training time

for smaller voice models from days to hours

30x improvement

in ML model training throughput

Terabytes of data generated

and stored weekly

Reduced carbon footprint

of video production



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Overview

Businesses looking to enrich their brand presence and communication with video content have traditionally faced steep production costs and a long process. Artificial intelligence (AI) technology startup [Synthesia](#) offers generative AI video creation as a service for customers to create realistic videos from text prompts in just minutes.

Synthesia wanted to make its video-generation software as a service (SaaS), Synthesia Studio, even faster by improving its machine learning (ML) models and upgrading its computing infrastructure for training these models. The fast-growing company needed a scalable infrastructure with high performance, so it used NVIDIA GPU-accelerated instances on Amazon Web Services (AWS) to significantly speed up and scale its ML training to support a user base growth of 456 percent.



Opportunity | Powering Fast AI Text-to-Video Generation Using NVIDIA GPU-Powered Solutions on AWS

Founded in 2017, Synthesia develops AI technology that customers can use to create instructional videos featuring stock or custom AI avatars. With Synthesia Studio, instead of waiting months for a full video production, companies can turn text into professional videos complete with natural AI voices in minutes. Synthesia supports the kind of rich storytelling and engaging communication that resonates with customers. It also prioritizes responsible use: all ML training data is sourced on the basis of consent, and all avatars are developed with consenting actors. In addition, the company is part of the Content Authenticity Initiative, which promotes the responsible use of synthetic media.



After growing rapidly to 350 employees and 50,000 customers, Synthesia found that training ML models on on-premises computers had become inefficient. In 2023, it decided to optimize its ML pipeline on AWS. Synthesia's production facilities generate many terabytes of data each week, and it requires high compute capacity to train its text-to-video ML models. With more than 50 ML researchers training large models, the company needed a large, scalable data lake and compute cluster to run multiple generative AI models 24/7.

Synthesia chose AWS because its fully managed services meet several critical requirements. "AWS has been embedded in our company from the beginning," says Jon Starck, CTO of Synthesia. "We aim to give more people access to video creation anywhere in the world, and we wanted to build for scale and self-service in the cloud. AWS provides a wealth of services that we wanted to adopt, and it has key initiatives for startups." For example, the company took advantage of the initial free compute capacity that AWS provides. The two teams also had regular meetings, and Synthesia saw the value in AWS support as it scaled up rapidly. "On AWS, we can scale and adapt our setup to meet our new needs," says Starck. "The AWS team provides the support to select the right services as well as financial planning to make it cost effective."

On AWS, Synthesia switched to multi-node compute clusters to do distributed ML model training on [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances powered by a variety of NVIDIA GPUs, using PyTorch. "Security, reliability, availability, and scale have all been critical as the company grows," says Starck. "We've had to adapt as we scaled. Access to managed services on AWS and the large number of services at our disposal have been hugely beneficial."

Solution | Accelerating ML Model Training by a Factor of 30 Using NVIDIA GPU-powered Amazon EC2 instances

The company adopted [Amazon EC2 P5 Instances](#) powered by NVIDIA H100 Tensor Core GPUs and [Amazon EC2 P4 Instances](#) powered by NVIDIA A100 Tensor Core GPUs accelerated model training by 30 times. It also uses [Amazon EC2 G5 Instances](#) powered by NVIDIA A10G Tensor Core GPUs for data processing and to optimize video rendering runtime. On AWS, Synthesia is generating customer videos quickly and providing previews faster than on its previous system. "We batch process datasets for new experiments, often running up to 100 instances at the same time," says Starck. "Experimentation is key to developing hundreds of avatars that deliver natural, lifelike performances and content."

The company manages its compute capacity using [Amazon EKS](#), [AWS ParallelCluster](#), and [AWS Batch](#). "We needed to train AI models and serve AI workloads at a large scale, and having these options for orchestration, management, and batching allows us to build the most optimal and efficient compute infrastructure for our use case" says Starck. Using these managed services and NVIDIA GPU-powered compute instances, Synthesia



reduced ML model training time for smaller voice models from days to hours. It's also storing its large datasets using [Amazon Simple Storage Service](#) (Amazon S3), object storage built to retrieve any amount of data from anywhere.

Synthesia builds its AI workflows on PyTorch, which is designed for flexible ML experimentation and efficient production. It also uses NVIDIA CUDA, a GPU-based parallel computing platform and programming model, for video rendering and inference.

Synthesia's user base has grown significantly in a short time, and the company is prioritizing sustainability as it grows. For example, the compute-heavy workloads involved in ML model training and inference run on NVIDIA GPUs in AWS Europe Regions, which are powered virtually 100 percent by renewable energy. Overall, in Synthesia's experience synthetic video creation is significantly more sustainable than traditional video production.

Outcome | Supporting Continued Growth with a Scalable Cloud Infrastructure

Synthesia is now working toward near real-time, interactive video rendering for its customers. As the company continues to grow, it plans to create even more advanced 3D avatars using large-scale 3D datasets stored in its data lake.

"We've been through a journey from local PC model training to large-scale training on multi-node compute clusters," says Starck. "On AWS, we've had the flexibility to change our infrastructure while maintaining the availability of compute to meet our workloads."

About Synthesia

Synthesia is an artificial intelligence (AI) technology company that develops a text-to-video software-as-a-service product for companies to generate instructional videos quickly using AI.

AWS Services Used

Amazon EC2 G5 Instances

Amazon EC2 G5 instances are the latest generation of NVIDIA GPU-based instances that can be used for a wide range of graphics-intensive and machine learning use cases.

[Learn more »](#)

Amazon EC2 P4 Instances



Amazon Elastic Compute Cloud (Amazon EC2) P4d instances deliver high performance for machine learning (ML) training and high performance computing (HPC) applications in the cloud.

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Amazon EC2 P5 Instances

Amazon Elastic Compute Cloud (Amazon EC2) P5 instances, powered by the latest NVIDIA H100 Tensor Core GPUs, deliver the highest performance in Amazon EC2 for deep learning (DL) and high performance computing (HPC) applications.

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Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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